

Data visualisation and exploration

Thomas de Jaeger
thomas.de-jaeger@epita.fr

Matplotlib & Seaborn: Visualization Essentials

Session 3

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Objectives:

- Learn to build basic and advanced plots in Python.
- Understand the structure of a Matplotlib figure.
- Use Seaborn for statistical visualization.
- Customize your plots for clarity and style.

Understanding Matplotlib

- Developed in 2003, inspired by MATLAB
- Forms the foundation of most Python visualizations
- Works well with NumPy, pandas, and SciPy
- Allows low-level control: ideal for scientific publications

Simple example:

```
import matplotlib.pyplot as plt
```

```
x = [1, 2, 3, 4]
```

```
y = [2, 5, 4, 8]
```

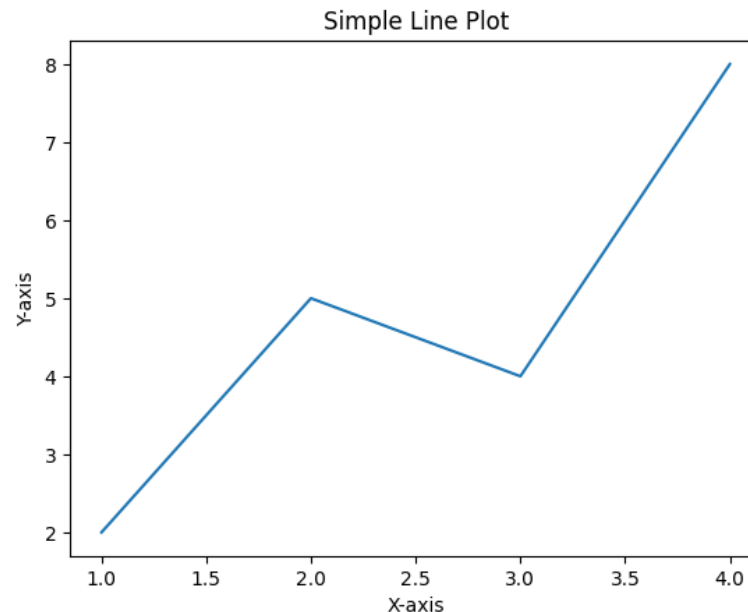
```
plt.plot(x, y)
```

```
plt.title("Simple Line Plot")
```

```
plt.xlabel("X-axis")
```

```
plt.ylabel("Y-axis")
```

```
plt.show()
```



Understanding Matplotlib

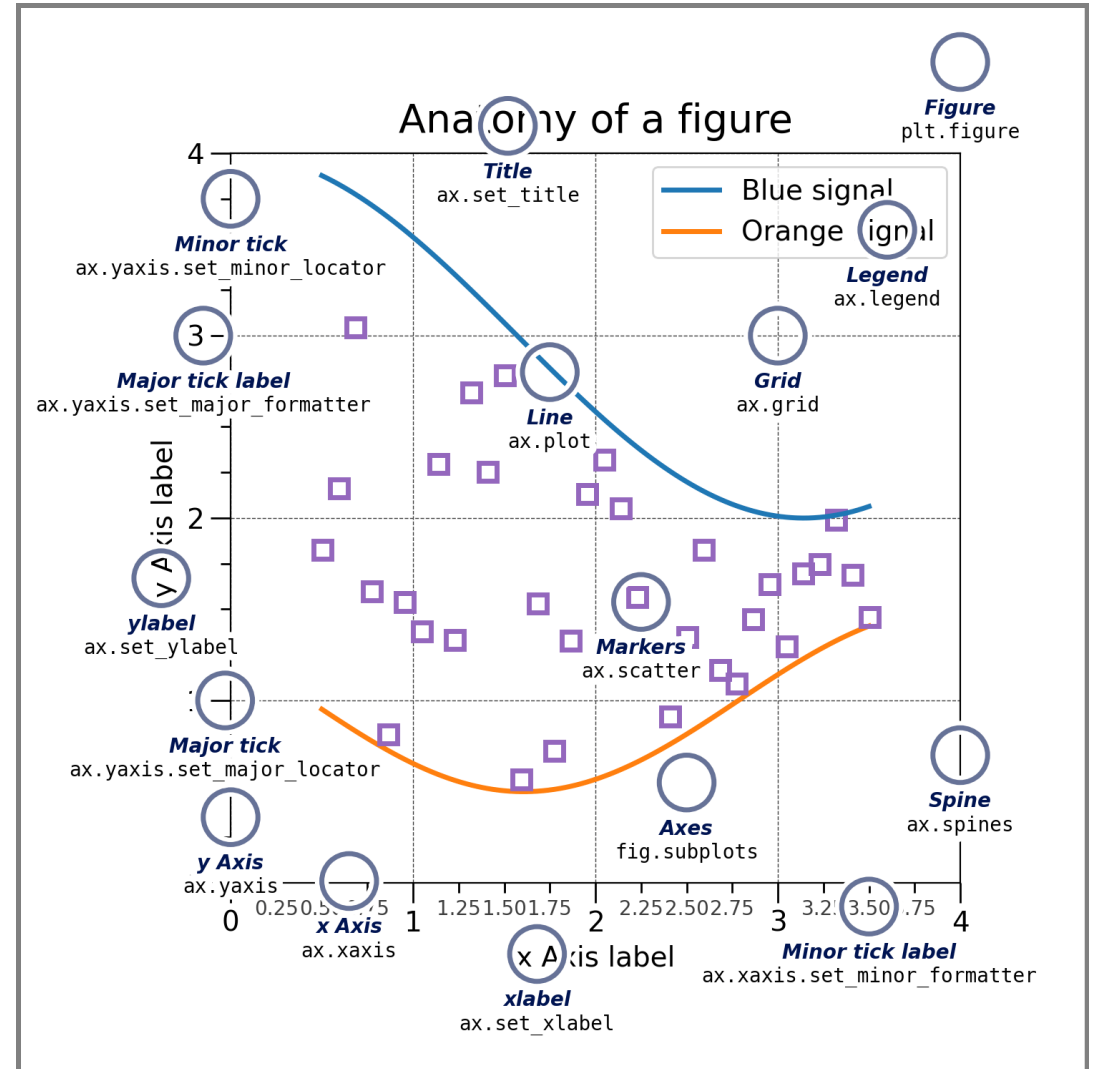
matplotlib primarily offers two levels of abstraction: the figure and the axes. The figure is essentially the “canvas” that contains one or more axes, where the charts are placed.

When you want to build a more complex graphs always use:

```
fig, ax = plt.subplots()
```

Ex:

```
ax.set_title("Monthly Sales", fontsize=16, color='navy')
ax.set_xlabel("Month", fontsize=14)
ax.set_ylabel("Revenue (€)", fontsize=14)
ax.legend(["Product A"], loc="upper left")
```



Set Global Plot Style with matplotlib.rcParams

rcParams (runtime configuration parameters) let you customize the default appearance of all plots.

When you want to build a more complex graphs always use:

Ex:

```
plt.rcParams['figure.figsize'] = (8, 5)      # Width, height
plt.rcParams['axes.titlesize'] = 16          # Title font size
plt.rcParams['axes.labelsize'] = 14         # Axis label size
plt.rcParams['axes.grid'] = True            # Turn on grid
plt.rcParams['grid.color'] = 'gray'         # Grid line color
plt.rcParams['lines.linewidth'] = 2         # Thicker lines
plt.rcParams['legend.loc'] = 'upper left'   # Default legend position
plt.rcParams['font.family'] = 'serif'      # Font family
x = [1, 2, 3, 4]
y = [2, 5, 4, 8]
plt.plot(x, y)
plt.title("Simple Line Plot")
plt.xlabel("X-axis")
plt.ylabel("Y-axis")
```

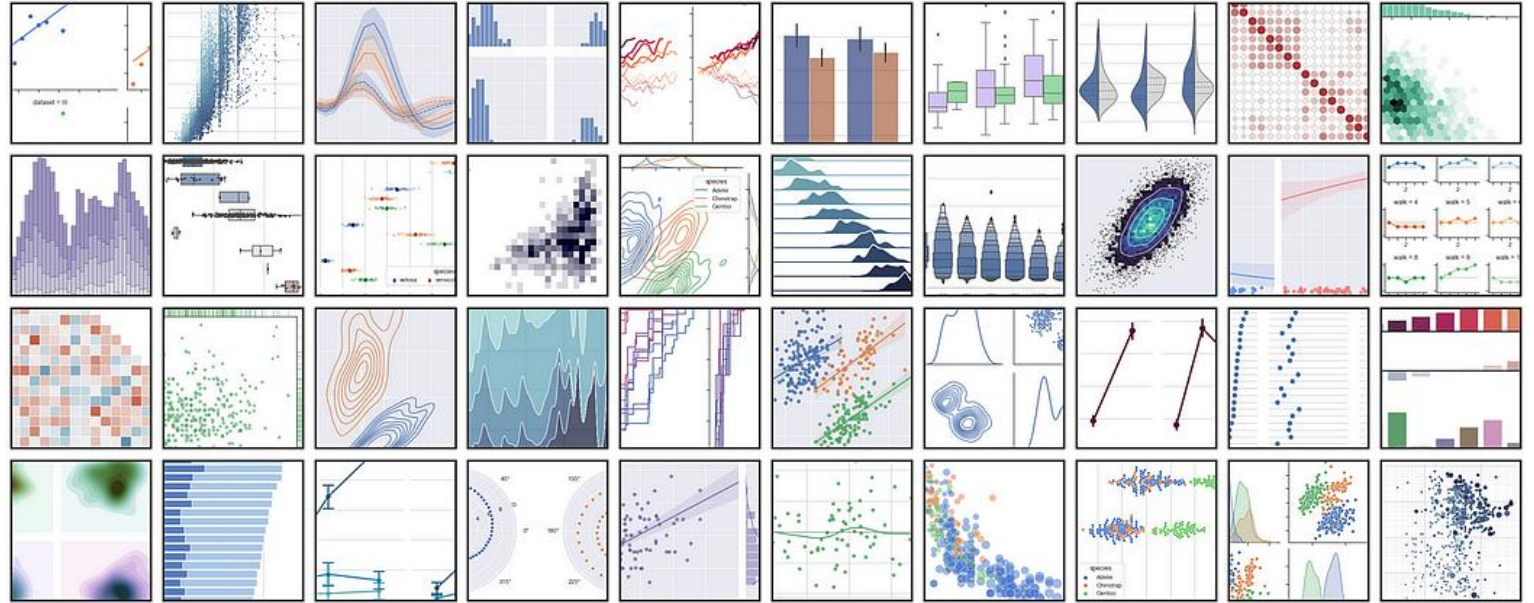
Category	rcParam Key	What it Controls	Example Value
Figure	'figure.figsize'	Default figure size (width, height)	(8, 5)
	'figure.dpi'	Resolution of figures in notebook	100, 300
Axes	'axes.titlesize'	Title font size	14, 'large'
	'axes.labelsize'	Axis label font size	12, 'medium'
	'axes.grid'	Whether gridlines are shown	True, False
	'axes.edgecolor'	Color of plot border	'black', 'gray'
Font & Text	'font.family'	Font style for all text	'sans-serif', 'serif'
	'font.size'	Global font size	12, 14
	'text.color'	Default text color	'black', 'darkblue'
Lines & Markers	'lines.linewidth'	Thickness of plotted lines	2.0
	'lines.linestyle'	Line style	'-', '--', ':'
	'lines.markersize'	Size of data point markers	6, 8
Ticks	'xtick.labelsize'	Font size of x-axis tick labels	10, 12
	'ytick.labelsize'	Font size of y-axis tick labels	10, 12
Saving Figures	'savefig.dpi'	DPI when saving a figure	300
	'savefig.bbox'	Bounding box behavior when saving	'tight'

Common Plot Types in Matplotlib

Plot Type	Function Example	Typical Use Case
Line Plot	<code>ax.plot(x, y)</code>	Trends over time or sequences
Bar Chart	<code>ax.bar(categories, values)</code>	Compare categories
Histogram	<code>ax.hist(data, bins=10)</code>	Distribution of continuous data
Scatter Plot	<code>ax.scatter(x, y)</code>	Relationship between two variables
Box Plot	<code>ax.boxplot(data)</code>	Spread, median, and outliers in distribution

What is seaborn?

- Built on Matplotlib
- Integrates with pandas DataFrames
- Automatic styling & color themes
- Built-in statistical functions
- Seaborn specializes in the art of statistical data visualization. It's not just about making data look good; it's about ensuring that data is presented with statistical integrity
- Seaborn arrives with a collection of eye-catching, pre-defined styles and color palettes.



Common Plot Types in Seaborn

Purpose	Function	Use for
Distribution	histplot()	Histograms (optionally with kde)
	kdeplot()	Smoothed distribution curve
Count Data	countplot()	Count per category
Relationships	scatterplot()	X–Y relationships
	lineplot()	Time series / trends
	regplot()	Regression line + scatter
Comparison	boxplot()	Spread, median, outliers
	violinplot()	Density + distribution + median
	barplot()	Mean (or other agg) per category
Heatmaps	heatmap()	Correlation or matrix data

```
sns.set_style("whitegrid") # "darkgrid", "whitegrid", "white", "dark", "ticks"  
sns.set_context("talk") # ("paper", "notebook", "talk", "poster« )
```

Common Plot Types in Seaborn

matplotlib

PROS

- Versatile and accessible
- Customizable
- Good documentation
- Universal data viz tool that plugs into many back ends

CONS

- Steep learning curve
- Users need to know Python
- Users need to understand the syntax of Matplotlib, which is based on the software, Matlab

seaborn

PROS

- Quickly creates simple visualizations
- Creates aesthetically pleasing visualizations

CONS

- Limitations on what you can customize
- Visualizations are not as interactive
- Users may need to simultaneously use Matplotlib